

Question 1 (15 Marks)

a) Let $z_1 = 1 - i$ and $z_2 = -1 + i\sqrt{3}$ 4

i) Find $z_1 z_2$ in the form $a + ib$.

ii) Find $|z_1 z_2|$ and $\text{Arg}(z_1 z_2)$. Hence write $z_1 z_2$ in modulus-argument form.

iii) Hence or otherwise show that $\cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$

b) z is a complex number such that $|z| = 1$. Using an Argand diagram or otherwise show that: 3

i) $1 \leq |z+2| \leq 3$

ii) $-\frac{\pi}{6} \leq \arg(z+2) \leq \frac{\pi}{6}$

c) If $1, \omega_1$ and ω_2 are the cube roots of unity, prove that 4

i) $\omega_1 = \overline{\omega_2} = \omega_2^2$

ii) $\omega_1 + \omega_2 = -1$

iii) $\omega_1 \omega_2 = 1$

d) Let $z = x + iy$ be any non-zero complex number. 4

i) Express $z + \frac{1}{z}$ in the form $a + ib$

ii) Given that $z + \frac{1}{z} = k$, where k is real show that $y = 0$ or $x^2 + y^2 = 1$

iii) Show also that if $y = 0$ then $|k| \geq 2$ and that if $x^2 + y^2 = 1$ then $|k| \leq 2$

Question 2 (15 Marks) *Start a new page*

a)

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i) Prove the identity $\cos^3 A - \frac{3}{4} \cos A = \frac{1}{4} \cos 3A$

ii) Show that $x = 2\sqrt{2} \cos A$ satisfies the equation $x^3 - 6x = -2$ provided $\cos 3A = -\frac{1}{2\sqrt{2}}$

★ iii) Find the three roots of the equation $x^3 - 6x + 2 = 0$, leaving your answers to four decimal places.

b) Let $x = \alpha$ be a root of the polynomial $P(x) = x^4 + Ax^3 + Bx^2 + Ax + 1$, where $(2+B)^2 \neq 4A^2$ 8

i) Show that α can not be 0, 1 or -1.

ii) Show that $x = \frac{1}{\alpha}$ is a root.

★ iii) Deduce that if α is a multiple root, then its multiplicity is 2 and $4B = 8 + A^2$

Question 3 (15 Marks) *Start a new page*

a)

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i) Show that $(x+2)$ is a factor of the polynomial $P(x) = 2x^3 + 19x^2 + 22x - 16$

ii) Find the zeros of the function $P(x) = 2x^3 + 19x^2 + 22x - 16$

b) Let $P(x)$ be a polynomial where $P(x) = x^3 - x^2 - 8x + 12$.

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i) Factorize $P(x)$, given that the equation $P(x) = 0$ has roots α , α and β

ii) Sketch the graph of $P(x)$.

iii) Let $Q(x)$ be another polynomial where $Q(x) = P(x)(x+a)$ and a is a constant chosen so that $Q(x) \geq 0$ for all x . Find the value(s) of a .

c) Using part (b) above or otherwise find the area under the curve $y = \frac{x^3 - x^2 - 8x + 13}{x - 2}$ from $x = 3$ to $x = 4$, leaving your answer correct to three significant figures.

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END OF PAPER

Question 1

a)

i) $z_1 z_2 = (\sqrt{3}-1) + i(\sqrt{3}+1)$ ①

ii) $|z_1| = \sqrt{2}, |z_2| = 2 \Rightarrow |z_1 z_2| = 2\sqrt{2}$ ①

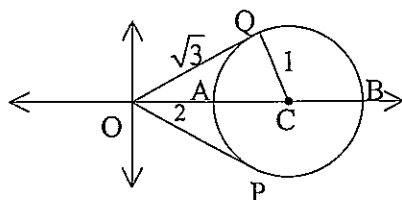
$\arg(z_1) = -\frac{\pi}{4}, \arg(z_2) = \frac{2\pi}{3} \Rightarrow \arg(z_1 z_2) = \frac{5\pi}{12}$ ①

$\therefore z_1 z_2 = 2\sqrt{2} \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$

iii) $2\sqrt{2} \cos \frac{5\pi}{12} = \sqrt{3}-1 \Rightarrow \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$ ①

b)

i) $z+2$ lies on a circle with centre $(2,0)$ and radius 1 ①



$1 = OA \leq |z+2| \leq OB = 3$

ii) $-\frac{\pi}{6} = \angle COQ \leq \arg(z+2) \leq \angle POQ = \frac{\pi}{6}$ ①

c)

i) wlog let $\omega_1 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$, $\omega_2 = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$

then $\overline{\omega_2} = \cos \frac{4\pi}{3} - i \sin \frac{4\pi}{3} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$

and $\omega_2^2 = \cos \frac{8\pi}{3} + i \sin \frac{8\pi}{3} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$ ①

ii) $1, \omega_1, \omega_2$ are the roots of $z^3 - 1 = 0$ ①

$\therefore$ the sum of the roots $= 1 + \omega_1 + \omega_2 = 0$ ①

iii) $\therefore$ the product of the roots $= \omega_1 \omega_2 = 1$ ①

d)

i) $z + \frac{1}{z} = x + iy + \frac{1}{x + iy} = x + iy + \frac{x - iy}{x^2 + y^2} = \left(x + \frac{x}{x^2 + y^2} \right) + i \left(y - \frac{y}{x^2 + y^2} \right)$ ①

ii) From i) $y - \frac{y}{x^2 + y^2} = 0 \Rightarrow y = 0$ or $x^2 + y^2 = 1$ ①

iii) if $y = 0$ then $x + \frac{1}{x} = k \Rightarrow x^2 - kx + 1 = 0$ has real solution $\Rightarrow \Delta \geq 0 \Rightarrow |k| \geq 2$ ①

if $x^2 + y^2 = 1$ then $2x = k$ and since $|x| \leq 1$ then $|k| \leq 2$ ①

Question 2

a)

$$i) \quad \cos 3A + i \sin 3A = (\cos A + i \sin A)^3 \quad ①$$

$$= \cos^3 A + 3i \cos^2 A \sin A - 3 \cos A \sin^2 A - i \sin^3 A$$

$$\therefore \cos 3A = \cos^3 A - 3 \cos A \sin^2 A$$

$$= \cos^3 A - 3 \cos A - 3 \cos^3 A \quad ①$$

$$\therefore \frac{1}{4} \cos 3A = \cos^3 A - \frac{3}{4} \cos A \quad ①$$

$$ii) \quad \text{If } x = 2\sqrt{2} \cos A \text{ satisfies } x^3 - 6x + 2 = 0 \text{ then,}$$

$$16\sqrt{2} \cos^3 A - 12\sqrt{2} \cos A = -2 \quad ①$$

$$\therefore \cos^3 A - \frac{3}{4} \cos A = -\frac{1}{8\sqrt{2}}$$

$$\therefore \cos 3A = -\frac{1}{2\sqrt{2}} \quad ①$$

$$iii) \quad \therefore 3A = \pm \cos^{-1} \left(-\frac{1}{2\sqrt{2}} \right) + 2n\pi \quad ①$$

$$\therefore A = 0.64405, 2.73845, 4.83284 \quad ①$$

$$\therefore x \approx 0.64405, 2.73845, 4.83284 \quad ①$$

b)

$$i) \quad P(0) = 1 \neq 0$$

$$P(1) = 2A + B + 2 \neq 0 \text{ as } 2+B \neq -2A$$

$$P(-1) = -2A + B + 2 \neq 0 \text{ as } 2+B \neq 2A \quad ①$$

$$ii) \quad P\left(\frac{1}{\alpha}\right) = \frac{1}{\alpha^4} + \frac{A}{\alpha^3} + \frac{B}{\alpha^2} + \frac{A}{\alpha} + 1 = \frac{1}{\alpha^4} (1 + A\alpha + B\alpha^2 + A\alpha^3 + \alpha^4) = \frac{1}{\alpha^4} P(\alpha) = 0 \quad ①$$

$$iii) \quad \text{Note that } \alpha \neq \frac{1}{\alpha} \text{ as } \alpha \neq 1, -1 \quad ①$$

$$\therefore P(x) = (x - \alpha)^2 \left(x - \frac{1}{\alpha} \right) (ax + b) \quad ①$$

$$\therefore a = 1 \text{ and } -\alpha b = 1 \Rightarrow b = -\frac{1}{\alpha} \quad ①$$

$$\therefore P(x) = (x - \alpha)^2 \left(x - \frac{1}{\alpha} \right)^2 \quad ①$$

$$\therefore P(x) = \left(x^2 - \left(\alpha + \frac{1}{\alpha} \right) x + 1 \right)^2$$

$$\therefore P(x) = x^4 - 2 \left(\alpha + \frac{1}{\alpha} \right) x^3 + \left[2 + \left(\alpha + \frac{1}{\alpha} \right)^2 \right] x^2 - 2 \left(\alpha + \frac{1}{\alpha} \right) x + 1 \quad ①$$

$$\therefore A = -2 \left(\alpha + \frac{1}{\alpha} \right), \quad B = 2 + \left(\alpha + \frac{1}{\alpha} \right)^2$$

$$\therefore 4B = 8 + A^2 \quad ①$$

Question 3

a)

i) $P(-2) = 0$

ii)

$$\begin{array}{r|rrrr} -2 & 2 & 19 & 22 & -16 \\ & & -4 & -30 & 16 \\ \hline & 2 & 15 & -8 & 0 \end{array}$$

$$\therefore P(x) = (x+2)(2x^2+15x-8) = (x+2)(2x-1)(x+8)$$

$$\therefore x = -2, \frac{1}{2}, -8$$

b)

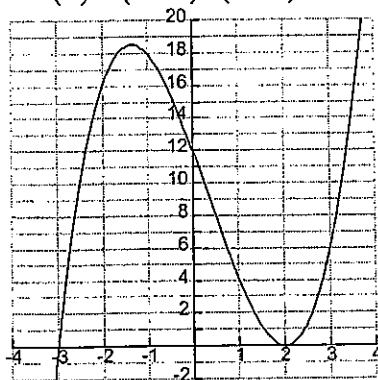
i) Let the roots be α, α, β

$$2\alpha + \beta = 1 \text{ and } \alpha^2 + 2\alpha\beta = -8$$

$$\alpha^2 + 2\alpha - 4\alpha^2 = -8 \Rightarrow \alpha = 2 \text{ or } -\frac{4}{3} \Rightarrow \beta = -3 \text{ or } \frac{11}{3}$$

$$\text{but } \alpha^2\beta = -12 \Rightarrow \alpha = 2, \beta = -3$$

$$\therefore P(x) = (x-2)^2(x+3)$$



ii)

iii) $Q(x) = (x-2)^2(x+3)(x+a) \geq 0 \quad \forall x$

$$\therefore x^2 + (3+a)x + 3a \geq 0$$

$$\therefore (3+a)^2 - 12a \leq 0 \Rightarrow (a-3)^2 \leq 0 \Rightarrow a = 3$$

c)
$$\int_3^4 \left(\frac{(x-2)^2(x+3)}{x-2} + \frac{1}{x-2} \right) dx = \left[\frac{x^3}{3} + \frac{x^2}{2} - 6x + \ln(x+2) \right]_3^4$$

$$= \frac{59}{6} + \ln 2 \approx 10.5$$